

Firmware Block Implementation for the Proto-DUNE II Hit Finding Architecture with the HLS Tool

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Abstract

The Deep Underground Neutrino Experiment (DUNE) is a large-scale international particle physics project based in North America. It will consist of two underground neutrino detectors placed within the path of an intense neutrino beam fired from Fermilab, the furthest of which will be 1300 kilometres from the source, in the Sanford Underground Research Facility (SURF). Objectives of the experiment include the study of certain properties of charge-parity violation (CPV) in the lepton sector, as well as the analysis of proton decay, and the investigation into black hole formation through supernova neutrino detection [1]. This report focusses on the ‘Trigger Primitive Generation’ (TPG) cores within the front-end electronics of the detectors. More specifically, the report discusses the hit finder architecture that processes optical data. This data is generated from argon scintillation within the ‘Time Projection Chambers’ (TPCs), as a result of neutrino-nucleon interactions. The project objective was to replace the existing firmware algorithms written for the detector’s ‘Field-Programmable Gate Array’ (FPGA) boards, with designs synthesised via Xilinx’s Vivado ‘High Level Synthesis’ (HLS) software. C++ code was used to implement the desired behaviour. One of the end goals was to allow the comparison of the new design’s performance and resource utilization to those written from scratch in the VHSIC Hardware Development Language (VHDL). HLS has the potential to make firmware design more accessible to a wider range of scientists. This could lead to more people contributing to improving the capability and performance of particle physics detectors. This would consequently enable the collection of groundbreaking data at a faster pace.

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1 Introduction

1.1 DUNE Experiment Goals and Context

The DUNE collaboration is an amalgamation of many neutrino-physics communities, centred around an opportunity provided by the large investment of the U.S. Department of Energy. This will contribute to a large expansion of the underground experimental architecture of the SURF facility, as well as a megawatt neutrino-beam facility positioned 1300km across North America at Fermilab. This investigation will be groundbreaking for several areas of physics, including neutrino oscillation studies, searches for proton decay, and for neutrino-based astrophysics i.e. black hole and supernova burst phenomena. Section 1.2 discusses the physics behind the experiment, and how DUNE could tackle key theoretical questions in the neutrino sector.

1.2 The Importance of Neutrinos in the Standard Model

The last particle predicted by the standard model (SM) to be experimentally confirmed was the Higgs boson. It was eventually discovered in 2012 by the ATLAS and CMS groups at the CERN Large Hadron Collider experiment in Geneva. This was a milestone of the predictive success achieved by the SM, and opened up a new saga in the exploration of physics beyond the standard model (BSM). There has not yet been success when searching for new BSM particles; in this way, the SM has agreed particularly well with experimental results.

The current formulation of the standard model was finalised in the 1970s, after the existence of quarks were experimentally confirmed [2]. Since then, all the remaining undiscovered particles predicted by the model have been confirmed, such as the top quark in 1995 [3], the tau neutrino in 2000 [4] and of course the Higgs boson [5]. The accurate prediction of weak interaction properties, as well as the properties of the W and Z bosons have further validated the SM. So why investigate BSM sectors of physics? There are still certain natural phenomena that remain to be unexplained such as the strong CP problem¹, neutrino oscillations (discussed in section 1.2.2) and the occurrence of dark matter and energy. Not only that, but the SM also fails to meet the requirement of being an all-encompassing, coherent theoretical framework of physics that links all phenomena and entities in the universe. These conditions are not satisfied because important issues such as gravitation, the accelerating expansion of the universe, or the asymmetry in the composition of matter and antimatter are not accounted for. The next question is: how can BSM physics be better understood?

Analysing the properties of neutrino mixing and mass could provide a strong opportunity to discover evidence for BSM physics. A wealth of data has been recorded since the dawn of the 21st century, kickstarted by the first observation

¹This is a contradiction between experiment, where CP symmetry is never violated by quantum chromodynamic interactions, and theory, where the current mathematical formulation of strong interactions state that it is possible for CP violation to occur.

of the disappearance of atmospheric muon neutrinos in 1998 [6, 7]. As a result, it has since been discovered that neutrinos oscillate in their flavour. A crucial conclusion can be made from this, which is the fact that neutrinos have mass, although unexpectedly small; the sum of the three neutrino mass eigenstates must be less than one millionth of an electron mass eigenstate [8, 9]. The origin of these small neutrino masses and their oscillations is still not confirmed however, and is up for discussion. DUNE exists to carry out an investigation of neutrino oscillations to test ‘Lepton-sector Charge Parity Violation’ (LCPV) as well as the ordering of neutrino masses and the three-neutrino paradigm². Before this is explored however, some preliminary concepts must be discussed.

1.2.1 Majorana versus Dirac Neutrinos

It is still not confirmed whether neutrinos are Majorana or Dirac particles. Majorana particles and their antiparticles are one and the same, whereas Dirac particles are not the same as their antiparticle counterparts. All standard model fermions besides the neutrino are known to be of the Dirac variation [11]. In the 1957 Wu experiment an observation of parity violation in cobalt-60 beta decay was made, leading to the conclusion that neutrinos must be ‘left-handed’ and antineutrinos must be ‘right-handed’ [12]. This is called helicity, where right and left-handed spins denote a parallel and antiparallel spin vector to the momentum vector, respectively.

This is not to be confused for a chiral phenomenon, where an entity is not identical to its mirror image. Chiral theories are those which are asymmetric with respect to chiralities i.e. does not respect parity symmetry. For massless particles chirality is the same as helicity. Massive particles on the other hand can have their helicity reversed by changing the frame of reference so that the momentum vector is moving opposite to the spin vector. It is not Lorentz invariant, and chirality and helicity must be distinguished. Therefore, the Wu experiment is not a confirmation that neutrinos are different to antineutrinos since their helicities can be reversed by changing the observational reference frame.

So what evidence is there to suggest that Majorana neutrinos exist? Firstly, the conservation of total lepton number could be analysed, where the total lepton number is given by $L = n_l - n_{\bar{l}} = L_e + L_\mu + L_\tau$, where n_l is the number of leptons, $n_{\bar{l}}$ is the number of anti-leptons and $L_{flavour}$ denotes the individual lepton number for a given flavour. Lepton flavour is not conserved in neutrino oscillation, but so far the SM supports total lepton number conservation. If there is an experimental observation of the nonconservation of the total lepton number due to the neutrino masses, then this would support the theory that neutrinos are Majorana type [13]. Majorana neutrinos would violate both total lepton number and baryon and lepton (B-L) conservation via neutrinoless double beta decay [14]. This interaction is only possible if the Majorana neutrino has at least one type with non-zero mass, the latter of which has already been

²The quantum mechanical mixing of the three neutrino mass eigenstates producing the three known neutrino flavour states [10]

established. Neutrinoless double beta decay is a process of great interest, being searched for by a significant number of experiments [15–18].

A substantial number of physicists believe that the matter-antimatter asymmetry is caused by chiral anomalies [19]. Chiral anomalies are theoretical instances of the nonconservation of a system’s total left and right handed chirality, and is a reason why the lepton number conservation law could be violated. These events are prohibited by B-L conservation, but this is contradicted by evidence of charge-parity violation showing that these events occur.

Neutrino oscillation is a mechanism within the electroweak model that makes it a chiral theory, and is discussed in section 1.2.2.

1.2.2 Neutrino Oscillation

Neutrino oscillation is the quantum mechanical transformation of a neutrino created with a specific lepton family number into a neutrino with a different family lepton number, which can be measured after some propagation through space. The probability of oscillation is proportional to the ratio of D/E , where D is the distance travelled and E is the energy of the neutrino [20]. The phenomena was experimentally discovered by the Super-Kamiokande and Sudbury neutrino observatories and was recognised by the 2015 Nobel prize for physics due to it being of great interest in the theoretical and experimental physics communities.

The oscillation phenomena appears due to the mixing of neutrino flavour and mass eigenstates. Previously, the neutrino state produced by weak decay was portrayed as a linear superposition of mass eigenstates with equal energy or equal momenta. This was corrected by describing the neutrino as entangled with the other particles emerging from the decay. For practical purposes, one can describing a travelling neutrino state as a mass eigenstate [21]. As the neutrino superposition of the mass eigenstates propagate through space, the quantum mechanical (QM) phases of each eigenstate change at slightly different rates, due to their mass differences. A neutrino born as one species will over time become a superposed mixture of e , μ and τ neutrino states. The mass differences between these flavours are small compared to their long coherence lengths i.e. the distance at which a quantum mechanical wavefunction maintains a specified degree of coherence. Macroscopic travel distances are required to observe this oscillation effect [22], such as the long baseline of the DUNE experiment. Section 1.2.3 discusses one of the objectives of DUNE: searching for the mechanism responsible for the mass of neutrinos.

1.2.3 Neutrino Mass Origins and Sterile Neutrinos

It is important to note that many investigations into neutrino physics involve BSM concepts i.e. contradictory to the SM. Right handed neutrino singlets are not predicted by the SM and the SM Lagrangian is renormalizable i.e. dimensionless when integrated [13]. These properties would suggest neutrinos are massless, yet as mentioned before, the experimental observation of neutrino oscillation contradicts this. It is not yet sure how neutrino masses arise. In the

SM, fermions have mass due to interactions with the Higgs field, which involved left and right-handed versions of each fermion. This means that sterile neutrinos are theoretically well motivated, since all fermions have been observed with left and right chirality [23].

The huge disparity in mass scales between neutrinos and the rest of fermions remains to be explained. A possible solution to this issue could be the existence of a sterile (right-handed) neutrino, yet only left-handed neutrinos have been observed so far. The solution includes a Majorana mass term for right handed neutrinos, that in turn interact with the left-handed neutrinos in the same way that the Higgs field interacts with the rest of the SM fermions. Right-handed neutrinos are predicted to be heavy since they would induce a very small mass for the left-handed neutrinos. ‘Active’ (left-handed) neutrinos would be proportional to the reciprocal of the heavy mass. The right-handed neutrinos would have to possess an energy close to the GUT scale (of order 10^{16} GeV) to reflect the neutrino mass limits that have been determined. This is called the ‘Seesaw mechanism’.

The term ‘sterile neutrino’ is used to differentiate right-handed neutrinos (or left-handed antineutrinos) from the SM active neutrinos with isospin charge of $\pm\frac{1}{2}$, therefore weakly interacting. The sterile kind only interact via Yukawa interactions with ordinary leptons and Higgs bosons, which means they influence and are influenced by the gravitational field and can mix with ordinary neutrinos through the Higgs mechanism³. This makes them promising candidates for dark matter [23]. Linking back to section 1.2.1 one can associate sterile neutrinos with the underlying question: do neutrinos and antineutrinos differ only in chirality, or do sterile neutrinos exist as separate particles from the common left-handed neutrinos and right-handed antineutrinos?

Now that sufficient context has been given, these concepts will be tied into one of the most important searches in BSM physics.

1.3 Charge Parity Violation and Baryon Asymmetry

Cosmology tells us that at the beginning of the universe, there existed equal amounts of matter and antimatter. This contrasts the current asymmetrical composition, suggesting there must be differences in the behaviour between baryonic and antibaryonic matter i.e. the symmetry of charge conjugation $\times$ parity must be violated. This means that in a system, if all particles were replaced by their antiparticles and their coordinates were ‘inverted’ by a parity operator, then the behaviour of the system would change. One of the first instances of observed CP violation is the decay of a neutral kaon to two pions [24]. Kaons have been observed mixing between the particle and antiparticle versions of themselves via CP-invariant oscillations. The decay of a Kaon into pions with $C \times P = \pm 1$ meant that the initial and final states were different. This was later understood in terms of the dynamics of quarks.

³The spontaneous breaking of electroweak symmetry during interactions, causing the bosons interacting with the Higgs field to have mass, when they otherwise would not.

$$\begin{bmatrix} d' \\ s' \\ b' \end{bmatrix} = \begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} \cdot \begin{bmatrix} d \\ s \\ b \end{bmatrix}$$

Figure 1: CKM matrix equation

This equation demonstrates the conversion of the weak interaction doublet partners of negatively charged quarks into the mass eigenstates of the original negatively charged quarks via the CKM matrix. d , s and b denote the down, strange and bottom quarks respectively. These are referred to as ‘down-type’ quarks. Similarly, u , c and t denote the up, charm and top quarks respectively. They are referred to as ‘up-type’ quarks. Each coefficient V_{ij} represents the probability amplitude that the j th down-type quark decays into the i th up-type quark [26].

Since this observation, CP violation has been recorded in the strange, beauty and charm quark sectors. All conclusive data is consistent with the theory that CP violation is caused by a single complex phase in the unitary quark mixing matrix: the Cabibbo-Kobayashi-Mashawa (CKM) matrix. This matrix gives the amplitude for a negatively-charged quark to convert through a weak interaction into a positively charged quark. Another way of looking at it is each CKM element contains information on the strength of flavour-changing weak interactions involving freely-propagating quarks [25]. See figure 1 for the generalised version of the CKM conversion equation.

Four independent parameters are required to fully define the CKM matrix. One of the most common parameterizations is the Wolfenstein parameterization. Within this framework there exists two parameters ρ and η , whose magnitudes determine the complex phase of the matrix. They account for quark behaviour, including CP violating phenomena, with great success.

However, despite observing CP violation in QCD interactions, it would be incorrect to assume that baryon asymmetry has been accounted for. In actual fact, the current theoretical frameworks that link early-universe CP violation to present day quark CP violation yields an asymmetry too small compared to what is actually seen in the universe [27]. This would lead to the thinking that CP violation in the lepton sector may take responsibility for the dominance of baryonic matter. Leptogenesis is a scenario in which heavy neutral leptons, which are their own antiparticle, existed briefly before undergoing a CP-violating decay, and is a possible explanation for the observed composition of matter [28]. These heavy neutral leptons are predicted by the seesaw models mentioned in section 1.2.3.

To record the rate of CP violation in the lepton sector, one must search for the difference in behaviour between neutrinos and antineutrinos. The current approach is to compare the rates for two CP mirror-image processes, an example of which is detailed in figure 2. Not only is each particle converted to their antiparticle, but their chirality is also reversed. Regardless of whether neutrinos

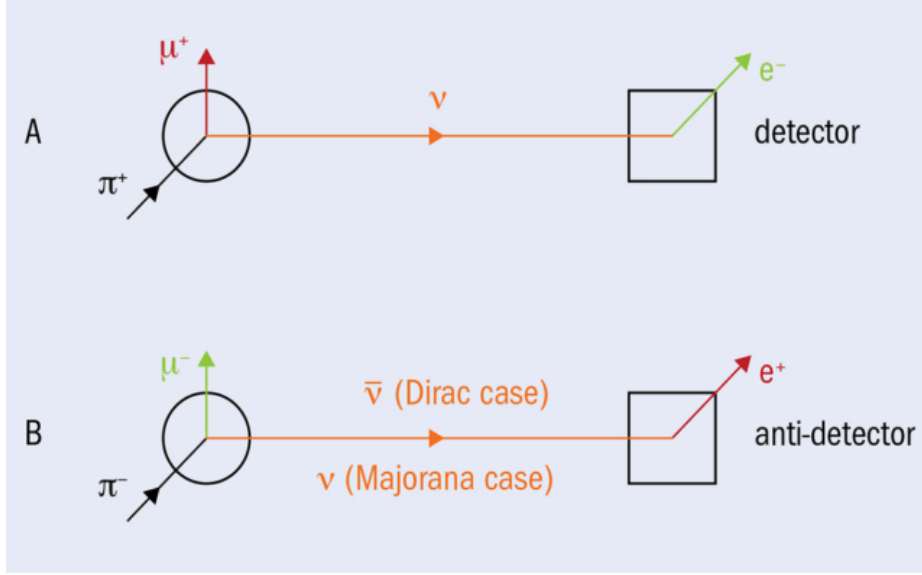


Figure 2: This figure shows the comparison of two CP mirror-image processes i.e. in process A, a π^+ decays into a μ^+ and a muon neutrino, that travels a macroscopic distance from the source to a detector, generating an electron from an interaction [29]. In process B, the exact same process occurs but each particle is replaced by its antiparticle. The ‘anti-detector’ in reality is impossible to implement, but the same detector can be used in both processes if the difference between the cross sections in process A and B are taken into account. The probability for the muon neutrino to produce an electron at the detector oscillates as a function of distance travelled over the particle’s energy.

$$\begin{bmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{bmatrix} = \begin{bmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu1} & U_{\mu2} & U_{\mu3} \\ U_{\tau1} & U_{\tau2} & U_{\tau3} \end{bmatrix} \cdot \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix}$$

Figure 3: PMNS matrix equation

This equation demonstrates the leptonic equivalent of the CKM matrix equation. The elements U_{l1}, U_{l2}, U_{l3} act as the coefficients for the eigenstate of flavour l to be given in terms of a linear combination of the mass eigenstates 1, 2 and 3 [31]. These define the PMNS matrix.

are Majorana or Dirac, if the rates are unequal between processes A and B, there is a CP invariance of the weak interactions of leptons.

Similarly to the fact that quark CP violation arises from a complex phase in the quark mixing CKM matrix, leptonic CP violation in neutrino oscillation can arise from the complex phase in the leptonic analogue: the unitary Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix. This matrix contains information on the mismatch of quantum states of neutrinos when they freely propagate over a distance and take part in weak interactions [30]. The PMNS matrix is part of a neutrino oscillation model. Why is the PMNS matrix important within oscillation physics and the DUNE experiment?

Neutrinos only interact via the SM charged-current and neutral-current weak interactions. The mass eigenstates can be defined as ν_1, ν_2, ν_3 with respective masses m_1, m_2, m_3 . As mentioned before, they are distinct from the charged-current interaction eigenstates (flavour eigenstates) which can be denoted as ν_e, ν_μ, ν_τ , labelled according to the charged-lepton that they couple with during a charged-current weak interaction⁴. The mass eigenstates can be expressed as linear combinations of the flavour eigenstates and vice versa [31]. This is shown in the eigenstate mixing equation in figure 3, where the coefficients of the linear combinations make up the elements of the PMNS matrix.

If the neutrinos are Dirac type, the PMNS matrix can be parameterized, like the CKM matrix, by three mixing angles $\sin^2(\theta_{12}) = \frac{|U_{e2}|^2}{1-|U_{e3}|^2}$, $\sin^2(\theta_{23}) = \frac{|U_{\mu3}|^2}{1-|U_{e3}|^2}$, $\sin^2(\theta_{13}) = |U_{e3}|^2$ and one complex phase $\delta_{CP} = -\arg(U_{e3})$. These parameters are being measured to greater precision in the DUNE experiment in order to better understand neutrino oscillation. δ_{CP} especially is not well known. Figure 4 visualises the probability of neutrino oscillation at the baseline of the DUNE experiment for different values of the complex phase. One can see that measuring the cross section statistics for this process at certain neutrino energies could give insight into the value of δ_{CP} .

If the neutrinos are Majorana type there would be two other physical complex phases. However, for practical purposes neutrino oscillation experiments cannot distinguish Majorana from Dirac neutrinos.

Current collaborations measuring neutrino oscillation and searching for lep-

⁴Mediated by the $W^\pm$ bosons.

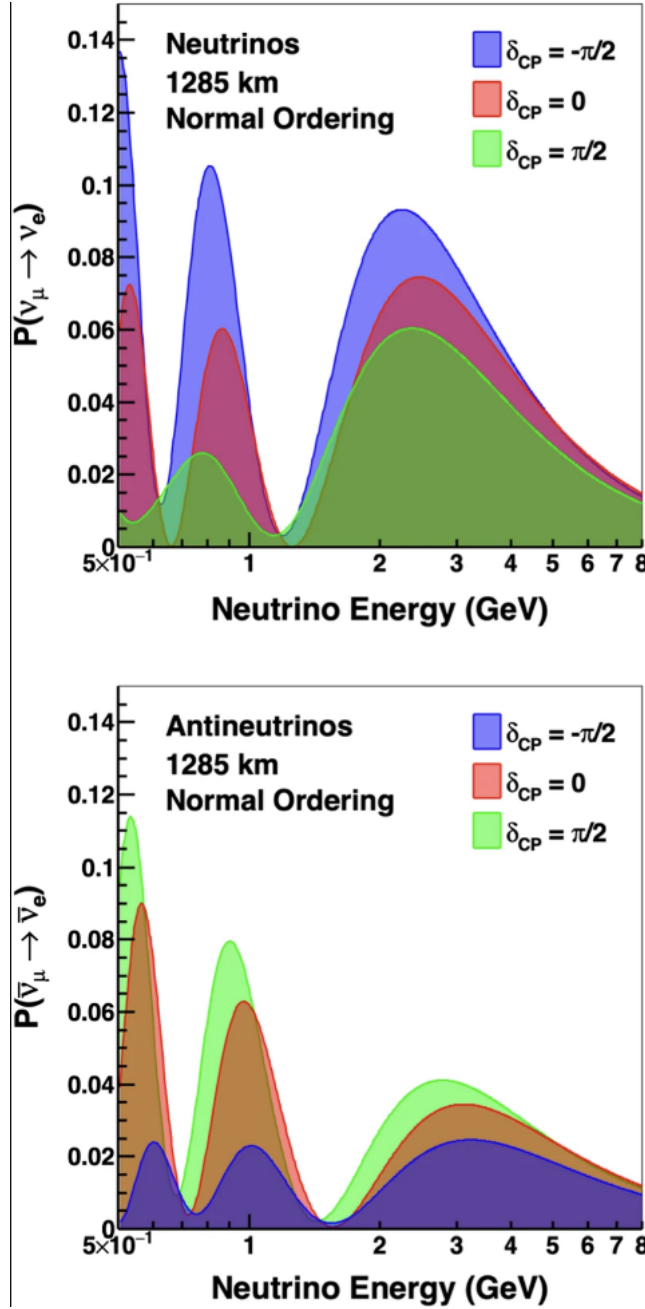


Figure 4: The probability that muon neutrinos (top) and antineutrinos (bottom) will appear at the far detector as electron neutrinos or antineutrinos respectively, depending on the complex phase value from the PMNS parameterization. The y-axis displays the appearance probability and the x-axis displays the neutrino energy. This graphic comes from the DUNE collaboration article featured in reference [32].

tonic CP violation include the NOvA and T2K experiments. The DUNE and Hyper-Kamiokande projects will continue to probe this concept with more sensitivity and precision. The next segment will explore how DUNE in particular will practically approach finding out if CP symmetry is unique to just quarks, or all constituents of matter.

1.4 An Overview of the DUNE Experiment

In order for DUNE to achieve its goals, three central components are required. The first is a high-intensity, neutrino sourced generated from a megawatt-class proton accelerator at Fermilab. The second is a large-scale underground far detector situated across the country at the SURF facility. Finally, the third is be a composite near detector installed 574 meters the neutrino source. To achieve the required precision on its measurements, the DUNE experiment will need to collect the data from several thousand neutrino interactions over the period of ten years. The number of interactions is influenced by the following:

- The intensity of the neutrino beam.
- The probability that the neutrino will oscillate (about 2%).
- The cross section of the interaction within the detector.
- The mass of the detector.

The highest power proton beam that a target can sustainably withstand is 1-2 MW, which practically limits the maximum neutrino intensity. Therefore the detector mass must be in the tens-of-kilotons range to achieve the required number of interactions. The cryostats stored within the far detector will hold approximately 17.5kt of liquid argon, reaching a total mass of 70kt per cryostat [33].

Not only must a huge mass of cryostats be used at the far detector, but they shall also be placed 1500m underground at SURF. This is because the rate of cosmic rays at the Earth's surface is 165 kHz, which would mean that 1 in 10^6 events would be due to neutrinos, which would decimate any opportunity to achieve the physics goals related to neutrino oscillation. At the planned depth of the far detector, the ratio of neutrino events to cosmic ray events would become slightly higher than 1, making the objective far more obtainable. If these requirements are met, how will the uncertainty of only a few percent be met when the far detector would only be able to measure the neutrino fluxes to the an uncertainty of 10% and the interaction rates to around 20%?

The near detector reduces these uncertainties by measuring neutrinos near the beam source in addition to the far detector data, so that physics measurements can be extracted from differences between the two. Another way to optimise the measurements is to place the far detector in the optimal distance range to determine the neutrino mass hierarchy, observe CPV and the complex phase parameter δ_{CP} . This is between 1000 and 2000 km, as shorter baselines

have lower optimal neutrino energies so that observables are too low energy to be visible. Charge parity sensitivity is also reduced by the ambiguity from unknown mass ordering. At longer baselines CP sensitivity is damaged by matter effects that increase proportionally to distance travelled by the neutrino [33].

1.5 ProtoDUNE DAQ System

DAQ architecture can be split into two parts: the front-end and the back-end. The front-end is comprised of circuitry that acquires the data stream from readout-electronics inside the detector, so that it can process and deliver the stream to the back-end: an offline storing, event-reconstruction computation farm. The focus of this report will be on the front-end, including the detection modules and the interface to the detector electronics.

There are several main factors that influence the design of the DUNE DAQ system. One of which is the hardware and maintenance cost per input channel, which is related to the technology life-cycle of each component. This contributes to the need for long-lasting essential parts such as the Xilinx⁵ FPGAs and network components. DUNE is set to run for approximately 20 years; the reliability and maintenance of the detector components is more important when compared to other large-scale projects. The power consumption of the detector modules also requires consideration due to cooling contingencies and the variance in running costs. Modularity of the hardware and firmware blocks is emphasised, so that one change in a sub-module would not affect the rest. This emphasis on customisability carries over to the HLS design process. If the design tools are more abstract and are rooted in commonly used languages such as C++, this will appeal to the vast majority of users (physicists)⁶ [34]. With these objectives in mind, R&D and fast firmware modifications can be a quick and efficient process. Section 1.5.1 will explore several types of detectors used in neutrino beam experiments.

1.5.1 Single and Dual Phase LAr TPCs

The use of TPC modules containing Liquid Argon (LAr TPCs) is a matured and popular technique used by several short-baseline⁷ neutrino experiments. They are popular for several reasons:

- Argon is a noble element of vanishing electronegativity, which means that electrons produced by ionizing radiation will not be absorbed as they drift towards the detector.
- Liquid Argon is inexpensive yet very dense, increasing the likelihood of a particle interacting by 1000 times when compared to Nygren’s TPC

⁵A United States based technology company that primarily supplies programmable logic devices and associated softwares.

⁶Discussed further in section 2.

⁷The baseline refers to the scale of the beam travel distance. DUNE is ‘long baseline’ as the detector is positioned across the continent from the beam source.

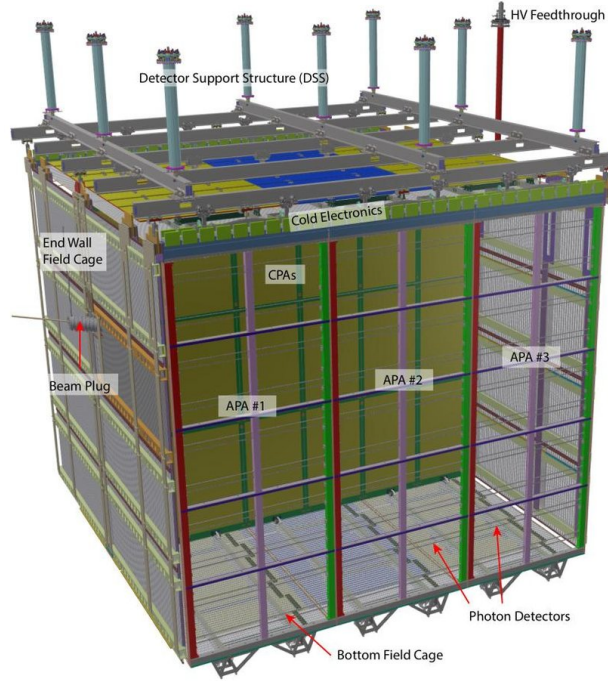


Figure 5: Diagram of the ProtoDUNE single-phase LAr TPC module. This contains two drift volumes, separated by a central cathode plane that is flanked by two anode planes, connected to a field cage that surrounds the entire module. Each anode plane is made of three Anode Plane Assemblies (APAs). These connect and send data to the front-end electronics being discussed in this report [35].

design [36]. This is important because the cross section for neutrino-nucleon interactions is very small.

- A neutrino-nucleon interaction causes the Argon to scintillate; a secondary energetic charged particle is generated and travels through the liquid, which causes the release of scintillation photons, the number of which is proportional to the energy deposited by the passing particle [37]. This contributes to the reconstruction of the event.
- LAr TPCs offer unprecedented 3D imaging capabilities and the functionality of a homogeneous calorimeter.

There exists both single and dual-phase versions of these LAr TPC modules. Inside single-phase (SP) liquid argon detectors, such as the structure featured in figure 5, the interactions are readout by wires immersed in the liquid and do not require the amplification of the ionized electron charge. The charge is localised using alternative induction and collection wire planes with different orientations. The charges drift horizontally via an induced electric field towards the vertically placed anode. The SP design has the disadvantage that its electronics must be especially low-noise due to the fact that no signal amplification takes place with the cryostat. These conditions were tested by ProtonDune-SP at CERN [35].

The dual phase (DP) detectors rely on the amplification of ionisation charges in the ultra-pure cold-argon vapour layer above the liquid. In DP LAr TPCs, ionisation charges produced by neutrino interactions drift vertically towards the liquid-vapour boundary where they are extracted into the gas phase, amplified by components named Large Electron Multipliers (LEMs), and collected by segmented anodes. Five photo-multiplier tubes are mounted underneath the TPC drift cage. They detect the Argon (prompt S1) scintillation photons formed from interactions with passing charged particles, as well as the (proportional S2) secondary scintillation light from the electroluminescence of the electrons extracted in the argon vapour. S1 is at least several orders of magnitude higher in detection amplitude than S2 in a tested TPC [38]. In this design, since the signal gain occurs in the gas phase, low-noise electronics are unnecessary.

For both designs, a field cage surrounds and defines the active LAr volumes, ensuring a uniform electric field throughout the cryostat. The DUNE collaboration plans on employing both SP and DP modules in the far detector, but the actual implementation will depend on the combined results from prototype detectors such as ProtoDUNE-SP and ProtonDUNE II.

In 2018 ProtoDUNE-SP collected hadron beam and cosmic ray data. When an interaction occurred in the LAr, the value for the total charge deposited along the track is converted to the energy lost from stopping the cosmic ray muons. Additionally, energy deposits were measured from artificially generated beam particles such as muons, pions, protons and positrons. This provided a unique set of high-quality data for the purpose of calibration and experimentalist studies. Hadron-argon cross section measurements are enabled by this new dataset, which is important for future analysis of neutrino oscillation phenomena within DUNE [33].

The next section will describe the interface system which handles the data acquired by the LAr TPCs.

1.6 FELIX

The LAr TPC modules used in the ProtoDUNE II design contain 150 APAs, each containing 2560 wires. There are 10 fibres/data streams in the APA, filled with 256 wires per fibre. The Front End LInk eXchange (FELIX) system is used to receive the data from the APAs, and was first developed within the ATLAS collaboration. Within this system contains the focus of this project: the TPG core, discussed in section 1.7. FELIX is a PC-based networking gateway with the FPGA mounted on PCIe⁸ Gen3 cards [39]. In ProtoDUNE II, FELIX interfaces with and manipulates the data from the APA, during which time it is fed through the TPG core within the Hit-Finder (HF) processor.

The readout window per neutrino event is between 3-5.5 ms for a 2.25 ms drift time. The FELIX architecture that handles this influx of events consists of 10 components called ‘MGT’ modules. The function of an MGT is to process the APA data stream into computational bits, which are then fed into 10 HF processors, as seen in figure 6. The hits are produced by processing the incoming ADC (Analog-to-Digital Converted) samples in parallel. The MGT output enters the HFs via the data router, which receives multiple-channel, single clock-tick packets. The packets initially exist in a format containing a given sample wave for all channels, and when processed by the data router are converted into a format subsisting of all the samples for a given channel i.e. multiple clock ticks of data. The formatted output is then transferred into 10 sets of 4 TPG block processors; one set for each MGT output. The processor outputs are multiplexed⁹ by the arbitrator into the Central Router (CR) interface [34, 40] as seen in figure 7. Section 1.7 will explain the TPG unit in further detail.

1.7 The TPG Core

Each TPG unit processes the data from 64 APA wires, and so there are 4 units for each core¹⁰. The TPG data consists of a header and a payload. The header determines the source wire, the time stamp, and the origin of the data. The payload is 64 bits of ADC sample data for hit finding. There is an AXI4-stream protocol¹¹ for the unpacking and reordering of data in the TPG block, featuring 6 general signals for various instructions in each component:

1. `tvalid` denotes if the bus data is valid for processing.

⁸High speed connection ports used in motherboards to connect GPUs, SSD’s, wifi chips and more.

⁹Combined into a singular signal output.

¹⁰ $4 \times 64 = 256$, the number of wires in a fiber; ten fibers summed over ten MGTs accounts for all wires within the APA (2560).

¹¹A standard interface to connect components that mutually wish to exchange data. The protocol supports multiple data streams that enable upsizing, downsizing and routing operations. The AXI4 protocol is part of an AXI family of micro controller buses, but this one in particular specialises in high performance, memory mapped component requirements.

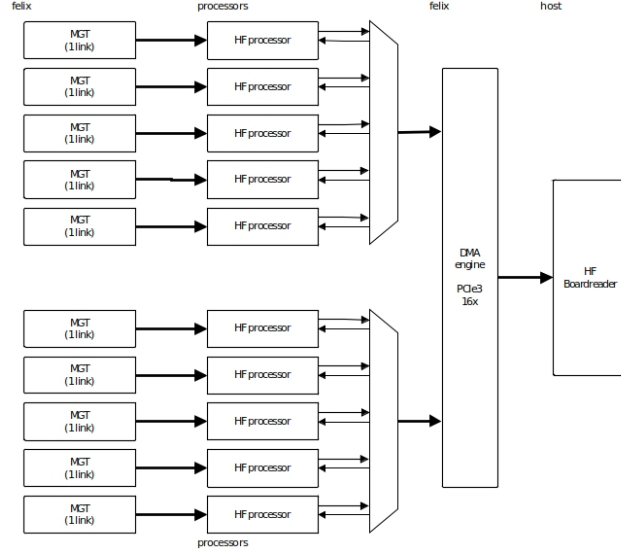


Figure 6: Overview of the Felix architecture in ProtoDUNE [41]

2. **tkeep** goes high when the incoming data should be processed.
3. **tuser** is a flag that denotes the end of a packet frame.
4. **tlast** indicates the end of a packet.
5. **tready** is used to apply back-pressure if the processing units ahead cannot process the data fast enough (all throughput is frozen).
6. **treset** goes high when the entire process must be reset and restarted, sometimes used when an error in the FELIX interface occurs.

Within the TPG core is a mechanism that strips the packet header: the Header Stripper/Combiner (HSC), featured in figure 8. The header is sent to the output of the block. Meanwhile, the remaining 64 ADC samples and their associated AXI4-S signals are processed by the PedSub (pedestal subtraction), FIR (finite impulse response), and HF sub-blocks in the chain. The data router is then signalled to send the next packet forward for input.

The objective of this project is to redesign the firmware algorithms for the TPG sub-blocks in Xilinx’s Vivado HLS software suite. This is to acquire a comparison of the performance and resource usage of the HLS design with the existing firmware written by VHDL engineers. Section 2 goes into more detail on this matter.

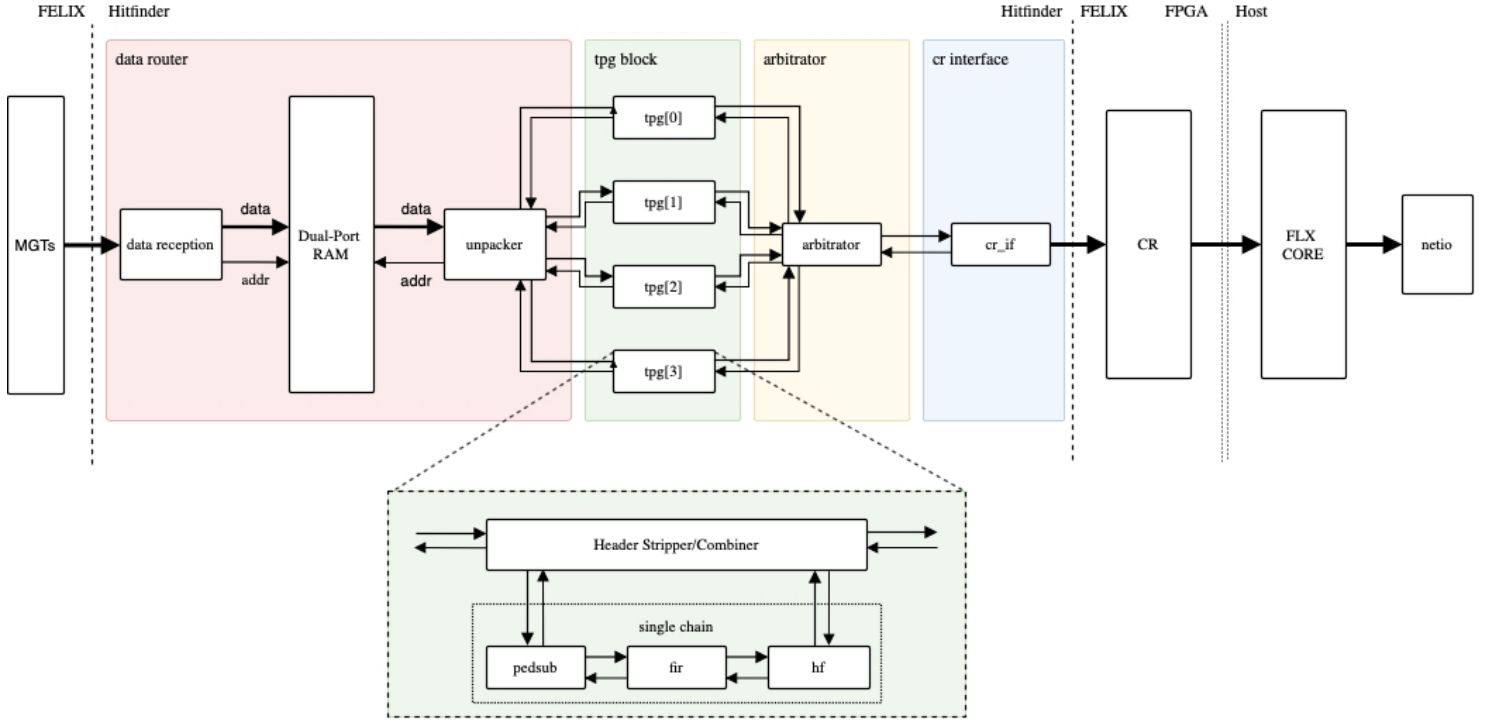


Figure 7: Hit Finder Firmware Architecture [41].

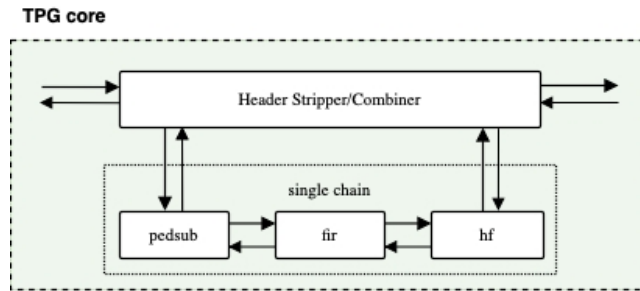


Figure 8: Focus on the HSC and the sub-block chain [41].

2 HLS

2.1 Background

High-level synthesis (HLS) is an automated design process that interprets an algorithmic description of a desired behavior and creates digital hardware that implements that behavior. It was initially designed to increase the levels of abstraction for design methodologies, forced by the growing capabilities of silicon technology and the increasing complexity of their applications. Beginning in the 1950s, assembly languages were introduced and gradually evolved to high level languages with improved compilation techniques, obeying the intuitive rules of grammar, syntax and semantics. Gate-level simulations appeared in the 1970s, and in the 1980s progress led to Hardware Description Languages (HDLs). A HDL is a programming language that describes the structure and behaviour of electronic circuits, and more commonly digital logic circuits [42]. HDLs look like programming languages such as C, but the main difference is that they include the notion of clock timing. The most commonly-used languages are Verilog and VHDL, the latter of which is used in the DUNE firmware framework. HLS allows for a connection between software languages and hardware implementation. In the case of section 3.1 this is achieved via VHDL firmware.

Synthesis begins with a high-level specification of the problem, where behavior is generally decoupled from low-level circuit mechanics such as clock-level timing. Early HLS explored a variety of input specification languages, although recent research and commercial applications generally accept synthesizable subsets of ANSI C, C++, SystemC and MATLAB. C++ was used to design the algorithms featured in section 3.1. This input source code is analysed, architecturally constrained, and scheduled to transcompile into a register-transfer level (RTL) design in a chosen HDL, which is in turn commonly synthesized to the gate level by the use of a logic synthesis tool [43].

RTL is a design abstraction which models a synchronous digital circuit in terms of the signal flow between registers and the logic operations being performed on said signals [44]. Hardware registers are circuits composed of flip flops (FFs): a sub-circuit that has two stable states used to store state information. They are a basic storage element in sequential logic, possessing the ability to read or write multiple bits at a time and use an address to select a particular register, used as an interface between the software and the computer peripherals [45]. RTL allows for circuitry simulation and digital analysis, and is used in HDLs to create high-level representations of a circuit, from which lower-level representations and ultimately actual wiring can be derived. RTL clock cycle data is given in Vivado HLS performance estimate data, as it can read how long a certain block, loop or function in the code took to complete; this is called the latency. Latency is a crucially important concept in RTL design, as it can make the difference between synthesizing a malfunctioning system or obtaining a synchronised circuit. Before the firmware block in question can be discussed, the design flow within Xilinx’s Vivado HLS development software will be explained to provide context for the generation of the algorithm.

2.2 Vivado HLS Design Flow

The design flow when approaching the synthesis of a C algorithm is as follows:

1. Compile, simulate, and debug the C algorithm.

This involves validating that the C program correctly implements the required processes. In order for this to occur, there must be a test bench written in C that includes the top-level function called within a simulation script, tasked with recreating the conditions of the implemented firmware, hosted within `main()`. This typically takes more time to design than the function to be synthesized.

2. Synthesize the C algorithm into an RTL implementation, optionally employing user optimization directives.

This allocates hardware resources such as Block Random Access Memory (BRAMs), data buses, look up tables (LUTs)¹², digital signal processors (DSPs)¹³ and registers. Each operation is scheduled to a certain clock cycle and bound to a functional unit within the board. Variables are bound to storage elements, and the RTL architecture is eventually generated [44]. See section 2.2.1 for more information on directives.

3. Generate comprehensive reports and analyse the design.

After C synthesis, output files are stored in a directory which contains folders for each RTL output format, and a report folder. The report folder contains files reporting on the performance and design corresponding to the top-level function and each individual subfunction. The analysis tab allows for cycle-by-cycle insight into the timing of logic operations and the FPGA resource consumption that they require. This is the best opportunity to analyse the effect of applied function directives and `pragma`¹⁴ commands.

4. Verify the RTL implementation using a pushbutton flow.

This is a post-synthesis validation process that compares the synthesised RTL and C programs to see if they generate the same output; this is called C/RTL co-simulation.

5. Package the RTL implementation into a selection of ‘Intellectual Property’ (IP) formats.

This generates the verified VHDL script that can be synthesized in the Vivado project manager into an implementable design. Before this can

¹²LUTs are arrays that replace runtime computation with a simpler array indexing operation, saving processing time [46]. FPGAs employ reconfigurable, hardware-implemented, lookup tables to provide programmable hardware functionality.

¹³DSPs are specialised microprocessing chips which are usually employed to measure or manipulate analogue signals. In the case of this project, they are used to perform rapid addition and multiplication calculations [47].

¹⁴Pragmas are directives that are stored within the source code, rather than in an external `.tcl` file

occur, a VHDL wrapper is written to connect the IP block’s input and output (IO) ports to the previous and following blocks, ensuring signals are being properly read and written. This is especially important for the AXI4 stream protocol discussed in section 1.7, as sufficient handshake and boolean signals must enter and leave the block for the AXI4 interface to be able to validate and manage the incoming and outgoing data samples. This can be tested in Questasim¹⁵ to provide a waveform analysis of every port and internal register involved. This is an important verification step in the design process, as the performance of an algorithm within the HLS environment can be vastly different to a VHDL simulation [43].

When writing HLS source code, there is punishment for using modern modules and storage entities; a simplified, classical style is most easily synthesized. An example of this is that storage arrays must have pre-allocated memory, with a determined number of elements at compile time (ruling out C++ vectors). Additionally, array manipulation modules such as `std::copy` and `memcpy` have very limited functionality or none at all, depending on the circumstance. These are some of many restrictions one faces when writing the desired behaviour in C++. Ultimately the code must be portable.

The performance of the logic operations come into question when constrained by a certain latency in order for the block to operate as required. The clock period must simultaneously stay below $1/f_{target}$, which in the case of the TPG core, the target frequency $f_{target} = 250$ MHz, meaning the clock duration must stay below 4ns. However, it must tick for long enough so that the block operations can occur within the desired latency. Performance comes at a trade to resource cost, as pipelining (parallelizing and rearranging operation initialisation to decrease the latency and input interval) employs more FPGA elements, which are a finite resource. One way to reduce this cost is to assign accurate bit-widths to variables. Using standard C data types for hardware design results in unnecessary resource usage. Operations can use more LUTs and registers than needed for the required accuracy, causing delays that may exceed a given clock cycle, increasing the overall latency. Smaller bit widths (e.g using 17 bit data types for 17 bit multiplication rather than the standard C 32 bit data type) result in hardware operators which are smaller and faster. More logic can then fit in the FPGA which can operate at a higher clock frequency. The next section will discuss the management of performance and area when choosing HLS design guidances.

2.2.1 Optimisations and Analysis

Directives act as a low-level guidance for the HLS tool, influencing the types of hardware used and the methods of implementing the source code operations [48]. A very useful example of this is the division of array elements into registers

¹⁵Software dedicated to simulating HDLs. This is typically used in collaboration with the Vivado project manager environment, where full VHDL block chains can be synthesized and implemented within an FPGA.

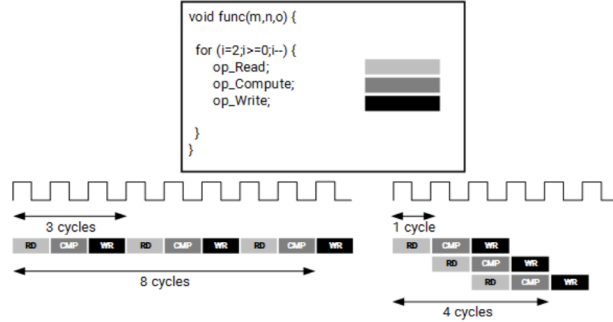


Figure 9: This figure contains an example showing how read, compute and write operations can happen in parallel to achieve higher throughput overall, using the `pipeline` directive.

using the `array_partition` directive, allowing for faster data access at the cost of increased resources.

Different optimization directives can be applied before C synthesis; one of the most crucial is the `pipeline` directive. This instruction forces code tasks to execute non-sequentially, allowing the next execution of a task to begin before the current one finishes, visualised in figure 9. One can infer from the figure that dependencies between two different operations can affect software’s capability to pipeline the entire function, meaning the initiation interval¹⁶ (II) cannot equal one until code changes occur and dependency directives are implemented. It can be deduced that if back pressure went high during the operation of the pipelined function `func(m,n,o)`, 3 cycles (the number of active function ‘layers’ at any one time) would pass until the block is fully paused. This would not allow for an instantaneous response to a signal, if it was necessary. Hence the requirement for $II = \text{latency}$, which if we require a fully pipelined design, latency must equal one.

In order to develop a fully pipelined block, code changes must be made to remove internal dependencies such as multiple store and load operations on registers and BRAMs (or any storage resource) at very different times during the operation of the block. This can be remedied by reducing the number of times to which a register is written via logic branching. This can then be further improved upon by applying `dependence` directives that identify false dependencies between arrays or variables, and allow the function operation to be even more streamlined. The user must take caution with instructing HLS to recognise a false dependency for a variable where there could in fact exist a true dependency. This can lead to C/RTL co-simulation failure and block malfunction.

Functions called within the top function can employ the `inline` directive so as to reduce overhead and minimise pipelining issues when calling sub functions.

¹⁶The number of clock cycles before new inputs are processed.

Not only can performance be adjusted by directives, but the HLS tool can also specify a limit to the resources used by the block, allowing for a balance between speed and area usage. Additionally, more customisation opportunities arise from the ability to select a block’s IO protocol to make sure the final design can be connected to other hardware blocks with the same framework.

There are several viable ways to achieve an optimal directive configuration for user’s purposes, but the best way to gauge the effect of adding directives in the Vivado HLS tool is by analysing the cycle-by-cycle display of operations within the synthesis reports. One can analyse a number of aspects surrounding the performance estimates of the block, perhaps the most important of which is clock timing. This is the fastest achievable clock frequency before the block ceases to function. This must meet the target to synchronise with the surrounding firmware blocks.

Secondly, the block latency and II must be reviewed, since the internal operations of the block must have enough time to complete before the next input is processed. Without any pipeline directives, the latency defaults to one clock cycle less than the II so that this can be achieved. In many cases the top function logic and even the input data types must be rewritten several times before the design reaches its performance objectives. See 3.1.3 for an example of this.

Finally, the utilization estimates are key to making the trade-off between latency and FPGA areas. Limited budget and gate area means that unlimited resource usage is not an option, therefore this must be tracked.

Bear in mind that some of the observations made in this section are specific to the design of detector signal filtration algorithms, and is not necessarily optimal for all HLS design projects. There are many more directives and design techniques that can be found in the Xilinx documentation titled UG902 [43]. Now that a sufficient introduction into the design process has been given, the next section will explore the key work outlined in this report: the firmware algorithms employed by HSC block chain first mentioned in section 1.7.

3 Development and Findings

3.1 Pedestal Subtraction Algorithm

The pedestal subtraction algorithm was the first mechanism in the project to be redesigned in HLS. It removes each channel’s base background noise from the incoming event data, so that the following FIR and HF blocks are dealing with the isolated optical signals from the neutrino events occurring within the detector. This should not be confused with the reduction of random noise of the isolated signal, which is taken care of later in the block chain. It is implemented within first sub-block in the TPG core, `pedsub`, which is fed by 64 sample words¹⁷ from the Header-Stripper/Combiner shown in figure 8. Before

¹⁷Known as a data packet, each sample contains an input ADC value and the associated AXI4 signals.

the performance and resource utilization can be compared to the original design, it is necessary to demonstrate the algorithm’s mechanisms.

3.1.1 Algorithm Operations

The input packet initially contains 70 sample words. However, the first 6 16b words are stripped from the incoming samples by the HSC and the remaining sample are sent to the `pedsb` block. There are three important variables used to achieve the reduction of background noise: the pedestal (or median), the accumulator, and the incoming ADC sample. The pedestal acts as the average background signal for a given channel, used to subtract from the data samples to produce the ‘true’ isolated signals. The pedestal and accumulator are adjustable values that must be set to an estimate before any samples have entered the block. When the pedestal is greater than the incoming ADC value, the accumulator is decreased by one, and if the opposite is true, the accumulator is increased by one. When the accumulator reaches a limit of ± 10 , the pedestal is increased by one or decreased by one respectively, and the accumulator is reset. See figure 10 for an illustration of how the algorithm manages background signal noise.

Connected to this block, or implemented within it, is a State Save/Restore (SSR) mechanism, which is used to save these adjustable variables at the end of a packet. For the HLS design that relies on an external SSR mechanism, this is implemented via a VHDL source file connected to the `pedsb` wrapper, that receives and feeds values to the pedestal subtraction logic. This can also be managed internally, the method for which is discussed in section 3.1.3.

The SSR functionality enables a variable restoration every time a packet enters a previously-processed channel, allowing for two constantly fine-tuned arrays of median and accumulator values to be used for the remainder of the detector run time. Therefore once `tlast` goes high, the final pedestal and accumulator values for the current channel are saved and those stored for the next channel are retrieved from the SSR. Every 64 packets, a saved value is again restored for the next packet entering the same channel. This is because packets are processed for each channel sequentially, beginning from 0 and reaching channel 63. This means that once the `pedsb` block has run for every channel, there will be a better background noise estimate and therefore more accurate ADC output data from the pedestal subtraction. Eventually, with a large enough flux of packets, one would obtain an optimal pedestal value appropriate for the background noise of each input channel.

3.1.2 Redesign of Pedestal Subtraction with External SSR

The development and debugging of the `pedsb_HLS` block led to a number of insights for future use of high-level synthesis when designing detector firmware. The first thing to note is that when developing an algorithm, it is advisable to start with a simplified version of the target function. This is because complex functions are especially prone to invalid signals and incomplete output, which can be difficult to debug when trying to trace back the unfamiliar proxy signals

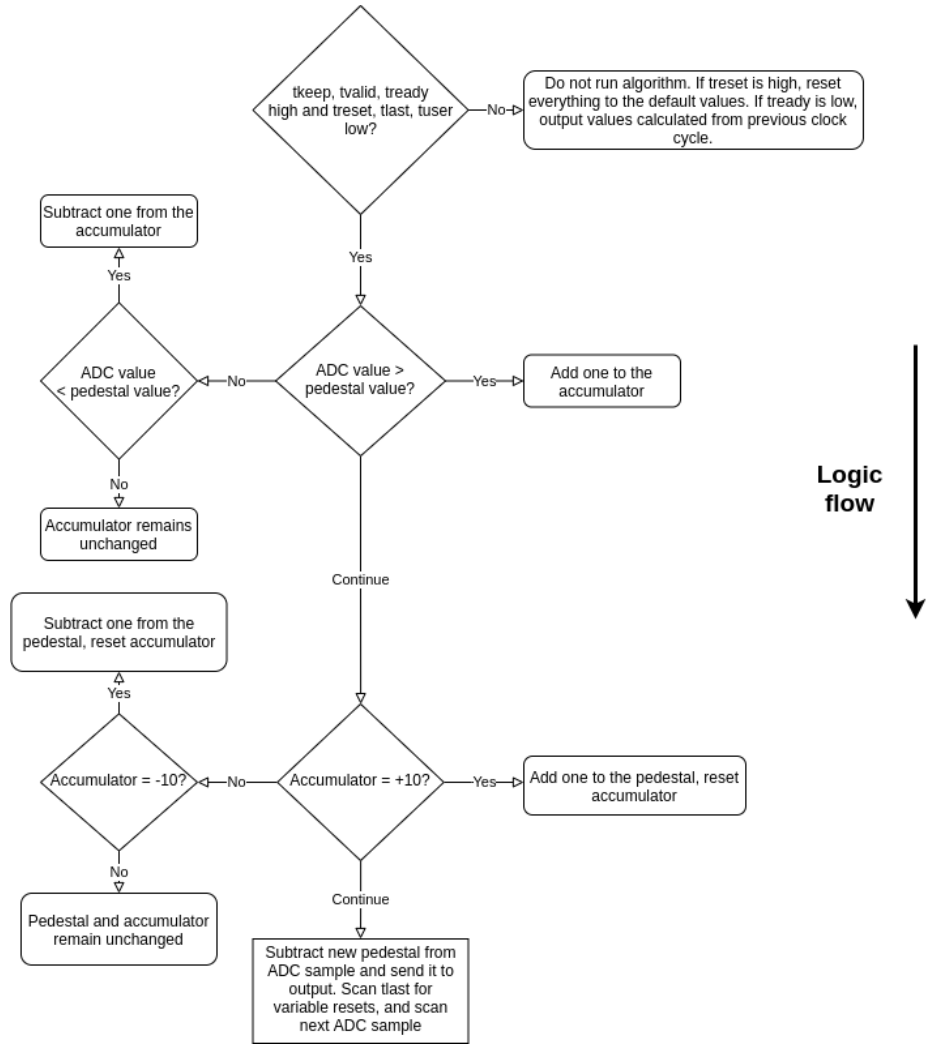


Figure 10: Flow chart detailing the logic flow of the pedestal subtraction algorithm, demonstrating the adjustment of the pedestal and accumulator values, as well as the occurrence of the subtraction from the ADC value.

in machine-written VHDL code.

One could argue that it is better to fix output issues within the HLS environment rather than try to treat ‘symptoms’ of the problem by editing the IP block file. There are limited Questasim warnings when building a simulation project to test the generated HLS IP, and observations can be made from the output signal waveform, but regular trial and error must be employed in order to target the source of an issue. This is a drawback of HLS IP, since the post-synthesis design lacks transparency when compared to writing HDL projects from scratch. One of the best approaches to deal with this is to slowly add layers of complexity, and each time an addition is made, one can test the functionality in Questasim and observe successful synthesis in the Vivado project manager before adding more features.

Contingencies must be included within the wrapper to account for a disparity between the IP block and wrapper port’s bits widths. Logic may also be required to compensate for the timing of handshake signals outputs, such as with the delayed internal variable restoration when `tlast` goes high, discussed below. This links into the inclusion of static variables that save internal values between function calls.

The Vivado HLS synthesis tool recognises a static variable in the same way a C++ compiler would, generating a static storage area for that value. An example of their use in the pedestal subtraction algorithm is the storage of a clock cycle counter that starts when the `tlast` signal output goes high. This static variable is saved between multiple function calls until it reaches three cycles. When this happens, the statically-stored pedestal and accumulator registers are restored to their SSR values for the next channel. Static variables are a crucial tool when a self-sufficient block is required and externalised IO management blocks are inconvenient.

However, one main issue faced earlier in the project was the long-term storage of the pedestal and accumulator: static variables were wiped between incoming packets. It can be seen from the `pedsub_HLS` Questasim waveform that this variable wipe occurs five clock cycles after the following packet’s `tvalid` signal goes high. A separate SSR storage array mechanism had to be developed for restoring values to a given channel in the long term¹⁸. In the preliminary report, HLS was deemed inconvenient when developing a long-term SSR mechanism; the block being called every clock cycle means that new declarations are reset very frequently, unless they are static. Further investigation needed to be made into the optimal approach, and this was found and utilised in the designs featured in sections 3.1.3 and 3.2.4.

Returning to the `tlast` mechanism, it can be discerned that the automatic static variable wipe happens too late; the restoration of values 5 clock cycles after `tvalid` goes high means that the first 5 subtractions occur using the previous packet’s pedestal, which leads to 5 invalid results for the next channel. A manual restoration of the internal values must occur before `tvalid` goes high

¹⁸i.e saving and then restoring packets 64 packets later, or a minimum of 4480 clocks after the initial packet enters a channel

for the following packet. However, this restoration must be after the previous packet's **tlast** output goes high; the **tlast** input is not used as this would still truncate the last value. This restoration window is prioritised in order to avoid invalidating the previous packet's ADC output values. It was found that restoring 3 clock cycles after the **tlast** output goes high allows for the restoration of the pedestal and accumulator to occur from the existing SSR mechanism. This is an example of how chronology can affect the output of a block in unpredictable ways.

Another clock timing concern is the required instant response to back pressure signal **tready**. Traditionally, the block would react too late as a result of the latency delay from input to output. This issue can be circumvented in the case that the block takes a maximum of 1 clock to complete, hence all functional designs mentioned in this report have a full operational time of 1 clock.

On input, the first logic branch depends on the **tready** input being high or low. If **tready** is low i.e. back pressure is active and the block operation requires pausing, the mechanism will continuously force an output of the same values that were sent through 1 clock cycle before the back pressure went active. This will cease only when **tready** goes high again. The desired behaviour can be achieved by constantly saving the previous clock's values and then restoring them to the output once the **tready** input goes low. However, if the block latency was greater than 1 clock cycle and the $II = 1$, back pressure would occur halfway through the operation of one function call, meaning the block would not react in time and erroneous outputs would be sent through to the FIR.

The final timing concern is ensuring that the outputs are transmitted at the same pace that the inputs are received, which can be achieved with the **pipeline** directive discussed in section 2.2.1 as well as careful optimisation of the logic. The next section will discuss the disparity in resource cost between the original design and the new HLS design.

Performance and Resource Utilization Comparison

The HLS block was forced to perform at a max latency of one clock cycle with an II equal to the latency, meaning the outputs arrive one cycle after the inputs and the inputs are accepted every clock cycle. This can be compared to the performance of the existing algorithm which has a latency of two and an II of one. The synthesis report shows the block has a clock cycle of 3.6 ± 0.5 nanoseconds, and an RTL co-simulation confirmed the block satisfies the target frequency of 250 MHz and a confirmed latency of one clock cycle. The HLS block outputs the same values as the previous design, with a 1 clock cycle delay due to the difference in latency. This was verified by a ModelSim simulation comparison between the blocks, and several model software output files.

When comparing resource utilisation between the **pedsub** blocks, detailed in figures 11 and 12, the most important components to compare are LUTs, FFs,

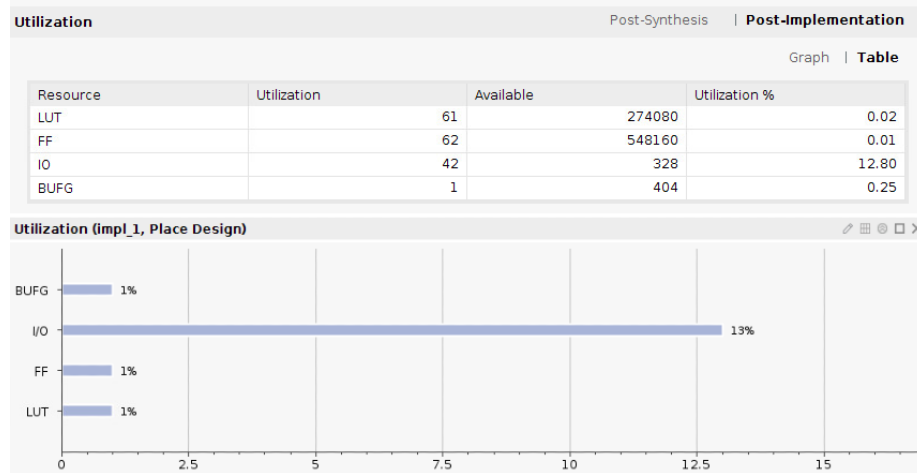


Figure 11: The resource utilisation for the original pedestal subtraction block including the exact figures above and percentage utilization below. All resource statistics were taken from the Vivado Project Manager post-implementation report summary for each block. Please note that for all resource bar charts, utilization percentages above 1% have been rounded to the nearest percent. Utilizations below 1% are rounded up to the nearest percent.

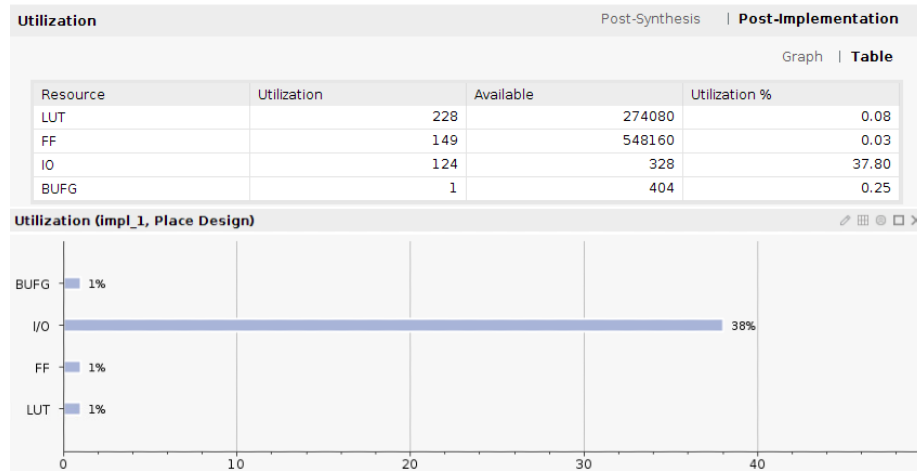


Figure 12: The resource utilization of the new pedestal subtraction HLS block connected to an external SSR, including exact figures in the statistics table above the graphical representation of the percentage utilizations.

and the input and output (IO)¹⁹ resources for the FPGA target²⁰.

The utilization tables show that the new design requires 0.06% more LUT usage when compared to the total LUTs available, which although comes to a 274% increase in total LUT usage for the block, is negligible when compared to the total available. Similarly, the FF usage for the HLS design shows an 140% increase from the original, but only a 0.02% difference in total usage of all available FFs. The disparity in LUT and FF usage can be explained by the additional internal logic and registers required to implement the delayed tlast restore mechanism and the instantaneous response to tready. The disparity is also derived from the halved latency of the HLS block when compared to the original block. This disparity could be dealt with by further investigation into the AXI4 Stream Protocol directives described in section 3.2.2.

Regardless, the difference in IO resource usage is not negligible, due to the large number of scalar and pointer type ports in the `pedsub_HLS` design compared to the original. There is a 25% increase in the FPGA target IO usage of the new design when compared to the total available, and a 195% total increase in the number of IO resources used. This can be remedied by removing unnecessary ports and incorporating multiple ports as members of a C++ `struct` type input, which can be manipulated further to reduce IO usage. This method is used in the designs featured in section 3.1.3, 3.2.3 and 3.2.4. The following segment will include discussion on the pedestal subtraction HLS block with built-in state save/restore behaviour.

3.1.3 Pedestal Subtraction with Implemented SSR

The pedestal subtraction algorithm `pedsub_HLS_SSR` is a HLS synthesized block including a memory-based SSR mechanism. The basic processing of the accumulator, pedestal and ADC values occur in exactly the same way as the design in section 3.1.2. However, in order to accommodate the performance deficit that came with storing and loading from a memory, the logic branching had to be adjusted. Additional temporary registers had to be utilised to avoid pipeline dependency issues. Arbitrary bit-width signed data types were used so as to reduce resource usage, and remove the need to truncate or zero pad input or output signals in the wrapper. Sub functions called within the top function were inlined and pipelined to reduce overhead. C++ `struct` data types were used as the function arguments (i.e. the block ports) with the original AXI4 stream signals as the struct members. For each new design in this project, a new testbench was written to validate the behaviour of the top function in the HLS environment. As previously mentioned, testbench scripts often took more time to write than the top functions themselves, but ultimately provided a priceless quality control for implementing new features.

¹⁹These are physical structures that allow the connection of an FPGA target to other devices within the system.

²⁰A programmable area or chip on the FPGA used to implement a digital circuit, in our case designed via HLS or directly through VHDL.

Memory Utilisation for the SSR Mechanism

The biggest change between `pedsub_HLS` and `pedsub_HLS_SSR` is the use of two global C++ arrays, defined outside the bounds of the top function, to save and restore the pedestal and accumulator values. These arrays can be accessed by the top function and act as permanent storage entities during runtime, solving the SSR problem faced in the preliminary report. HLS automatically implements these entities as two BRAM components. BRAMs support a very limited data access rate. This can cause problems during synthesis of the RTL files and needs to be worked around in some cases. The maximum access rate that a BRAM can support in this case is twice per clock cycle.

Fortunately, the only time `pedsub_HLS_SSR` needs to access each memory is to read and write once at the end of the `tlast` delay period. Therefore no pipelining or memory access issues were met at synthesis; the saving and restoring to the memory occurred seamlessly. However, in order to achieve latency = 1, these arrays had to be fully partitioned into registers to increase the speed that each memory array can be accessed. These registers were generated as ‘power-on initialisations’ similarly to static variables, except the partitioned global arrays are not wiped between packets. In examples such as the `fir_HLS_SSR` block discussed section 3.2.4, the memory is too large to be partitioned, and so BRAMs were necessary to store the SSR values whilst avoiding massive LUT and FF usage from full array partition. Different conditions call for varying memory implementations.

The `pedsub_HLS_SSR` function had to be streamlined extensively in order to fit all operations into one clock cycle, which as mentioned before is a necessary condition for successful back pressure behaviour generated internally, rather than via directives. In order to correctly save and restore the pedestal and accumulator, a static channel counter was used as a memory index between the values of 0 and 63. During the `tlast` delay period at the end of each packet, the current channel’s pedestal and accumulator values were saved to the SSR memory 2 clock cycles after `tlast` went high, and the next channel’s values were restored on the following clock cycle. This avoided pipelining dependency issues involving the partitioned arrays and the static accumulator and pedestal variables. The channel counter would also increase after the end of each packet.

The `treset` functionality was removed in order to provide more space for the SSR. This reset mechanism was proven obsolete since all storage entities were automatically wiped by the external FELIX system at the beginning of a data run. The only issue encountered in this case was making sure non zero starting values and booleans were set correctly at the start of a simulation, since these are prepared erroneously by the automatic restart. This can be achieved by declaring and defining global entities as their starting value on the same script line, which are then changed after the first clock cycle.

Performance and Resource Utilization Comparison

The main benefits of the redesigned pedestal subtraction block with an in-built SSR are increased performance (faster throughput) and the convenience of a self-sufficient block, with no external systems responsible for the memory access behaviour. The clock period minimum of this design was shorter than the previous algorithm, reported at $3.0 \pm 0.5\text{ns}$. The simulation output for multiple packet waves with a randomly oscillating back pressure signal matches the model output files used to test the original pedestal subtraction SSR block.

In contrast to the design in section 3.1.2, the IO resource utilization of the HLS block was more efficient than the from-scratch design, with a 16% decrease from the original design's IO usage and a significant 3.36% decrease when compared to the total IO resources available. These statistics can be found in figures 13 and 14. **Struct** type function arguments and more efficient port generation is responsible for this. However, the resource benefits end here.

The LUT usage of the HLS design is significantly higher at 481% of the original, or a non-negligible 0.26% usage increase of total LUTs available, which can be attributed to the LUTs required to process the operations on 128 registers for the pedestal and accumulator SSR arrays. FF usage increasing by 0.27% of the total amount available can also be attributed to the demanding SSR arrays. Area usage can be dramatically reduced by un-partitioning the SSR memories and pursuing the methods described in section 3.2.2 so as to avoid the demanding requirement for latency = $\Pi = 1$. This would use a small number of BRAMs instead of large numbers of LUTs and FFs, the analogue of which is seen in the original design requiring 32 LUTRAMs.

3.2 Finite Impulse Response Filter

The FIR algorithm acts as a random noise filter for the pedestal-subtracted ADC values, so that the final signal entering the hit finder block is clean enough not to generate false hits. See figure 15 for how the `pedsub_HLS_SSR` and `fir_HLS_SSR` ports connect. Included in figure 16 is a visual representation of the effect that the low-pass, 32-tap filter has on the incoming ADC values. From this figure one can see how the hits can be more easily identified by the hit finder block, since the peaks are more distinct and the time spent above a certain threshold value can be more accurately measured.

3.2.1 Algorithm Operations

The FIR filter's core mechanism involves accepting the same number of samples per packet as the pedestal subtraction algorithm, but the difference is that each incoming ADC value is stored in one of 32 'taps' (registers). A tap shift occurs every time a new input is processed, where the value at tap N moves to tap N+1. In other words, if we number the taps 0 to 31, and N new inputs have been processed since an input denoted X, then the input X will occupy the Nth

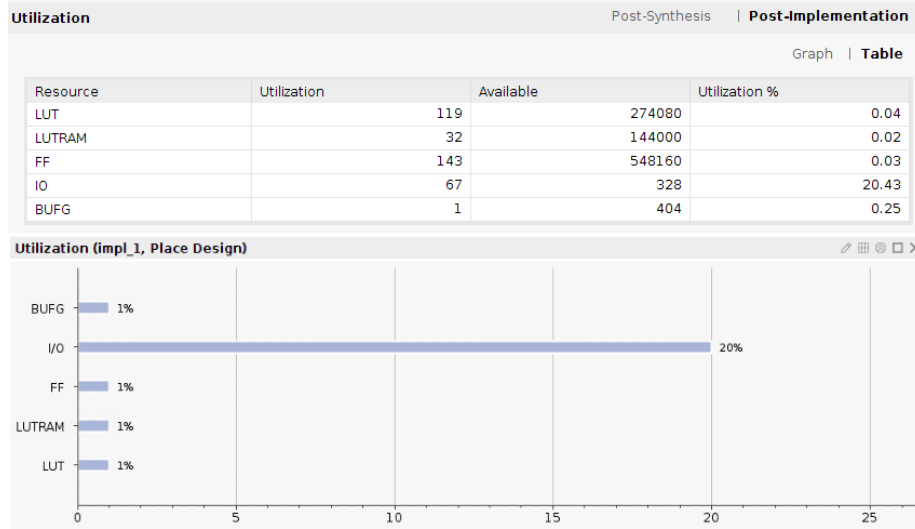


Figure 13: This figure details the resource statistics for the previously built pedestal subtraction algorithm, with the SSR mechanism included in the statistics. This is the result of the block being written from scratch in VHDL. This figure enables a comparison to the statistics of the `pedsub_HLS_SSR` block

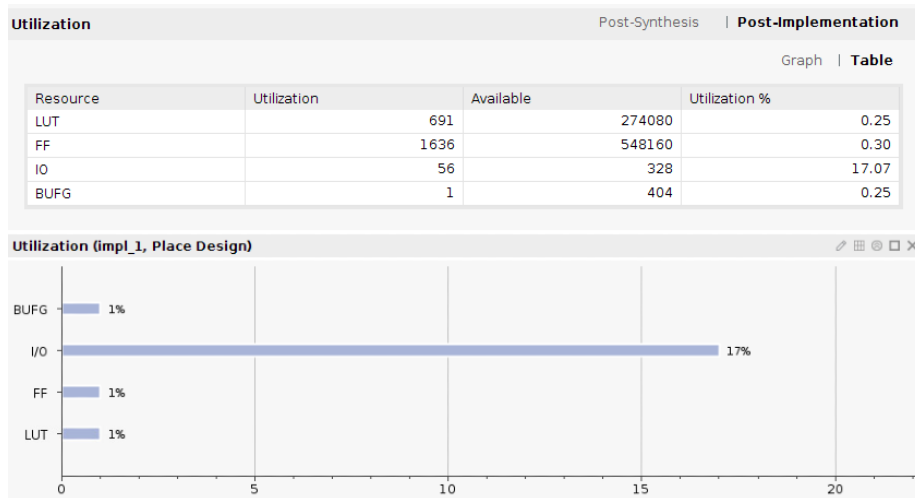


Figure 14: This figure details the resource statistics for the pedestal subtraction algorithm generated from HLS with an SSR memory built into the design, to be compared to the original `PedSub_SSR` block written directly in VHDL.

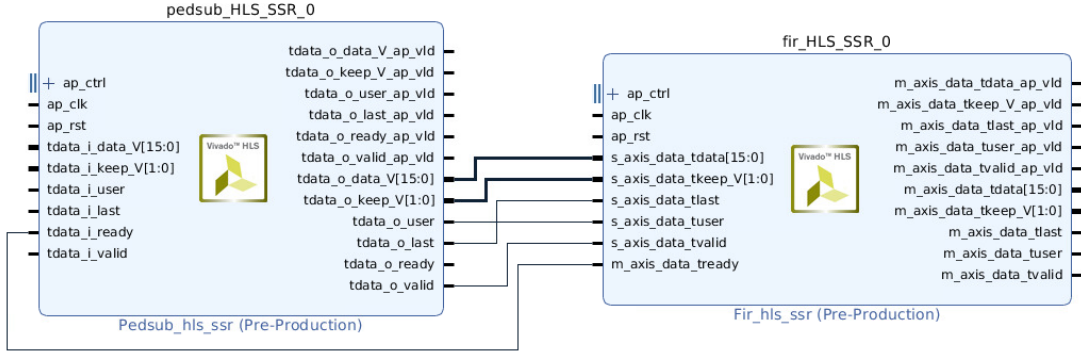


Figure 15: This illustration features the block diagrams of both the pedestal subtraction and FIR HLS IP blocks, and how their IO ports are connected (not including the HSC connection on the left side or the HF connection on the right side). Note `s.axis.data` signifies left-side ports and `m.axis.data` signifies right-side ports.

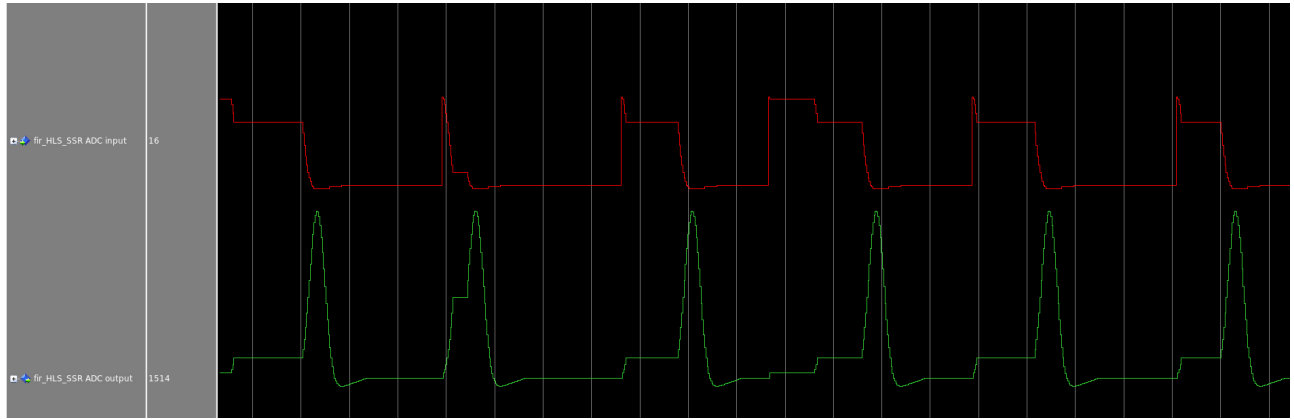


Figure 16: Detailed in the graphic are the analogue ADC input and output signals generated in a Questasim simulation. Note the clear peaks where the hits occur in the green output below, in contrast to the unfiltered red input above. Plateaus in the signal are a result of the random back pressure events pausing ADC input and output. This is not to scale; the green peaks are much larger than the red peaks in true scale.

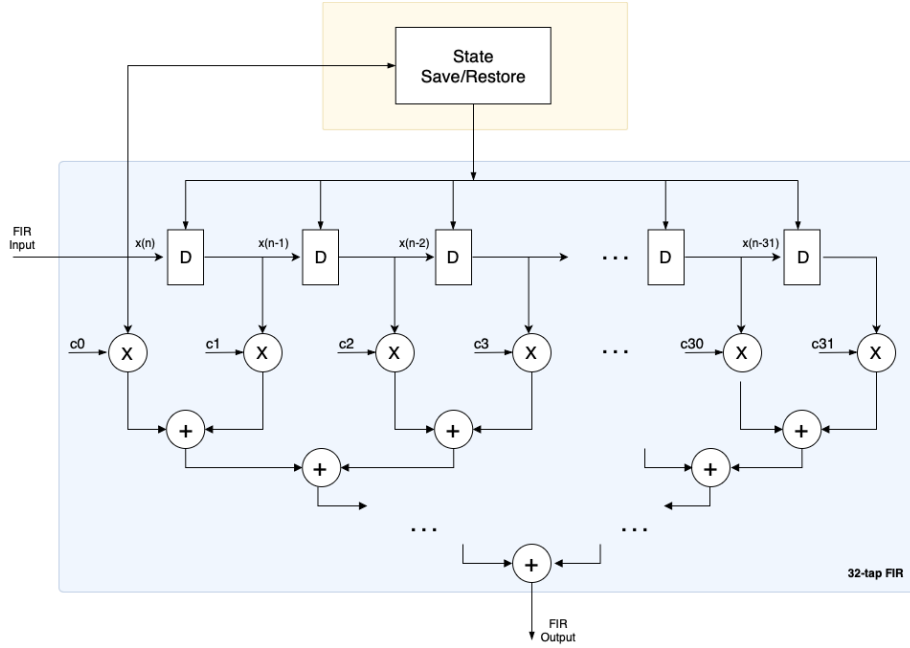


Figure 17: A visualisation of the FIR filter’s manipulation of incoming ADC values, including the register shift, coefficient multiplication, sum tree and the saving and restoring of the final 32 tap values [41].

tap along until it reaches the 31st, where it is forgotten on the next tap shift. Additionally, the final 32 inputs in the packet are saved to an SSR memory array for the channel being processed, and then will be restored to the taps next time that channel receives a packet, 64 processed packets later. This behaviour can be seen in figure 17.

The filter mechanism multiplies each tap by a certain customisable coefficient and sums the result of each multiplication to one value, which is sent to the ADC output. In figure 17 and in the original `FIR.StateSR` design, an addition tree is utilized to split the sums into 5 separate levels, the number of which that occurred were 16, 8, 4, 2 and 1 in that order. However, the approach taken in HLS was to sum each register individually in a loop, and then fully unroll that loop so all sums occur at the same time. This allowed for the necessarily high performance but required significantly more resources. Section 3.2.2 will discuss a possible workaround that would be investigated further if the project continued into the future.

3.2.2 Axis Directive Method for Back Pressure Behaviour

Initially, it proved difficult to achieve the performance required for an ‘instantaneous’ back pressure response in a pipelined FIR block. An alternative solution

was sought after, in order to achieve instantaneous back pressure for a block with latency greater than one clock cycle. A workaround was found, and it involved using `axis` interface directives on the input and output ports of the design. This enabled HLS to automatically implement the AXI4 stream protocol behaviour such as asynchronous back pressure from the `TVALID` and `TREADY` handshake detailed in Xilinx’s AXI reference guide. Side channel signals such as `TKEEP`, `TUSER` and `TLAST` can also be added as members of the port `structs`.

After correcting the FIR wrapper to accommodate the new HLS AXI stream ports, Questasim simulations demonstrated working back pressure behaviour for blocks of latency greater than 1 clock cycle, when a simplified version of the design is tested. The issue with this approach is that HLS struggles to fully pipeline designs with `axis` port registers, which is a minimum requirement for all designs used in the TPG core. Additionally, a bug was found that hard-wired `tuser` to high when `tready` output signal was sent to the wrapper’s output port (previously there had been no `tready` output signal from the HLS blocks). The combination of these two issues led to seeking a new fix as a consequence of time constraints. However, if more research was to be done, then this method would be investigated further due to it being a better general framework for meeting the needs of FELIX’s AXI specification system.

3.2.3 Redesign of FIR with External SSR

The HLS redesign of the original FIR block that relies on an external SSR is called `fir_HLS`, and when synthesized achieved a minimum clock frequency of $3.4 \pm 0.5\text{ns}$. It mimics the original algorithm, achieving the same output values for input files of 128 packets with random back pressure, but the methods to achieve these outputs differs slightly. This is mainly due to the accommodation of the one clock latency and $II = 1$ requirements. Mechanisms for `tlast` restore, back pressure logic and struct ports were reused from `pedsub_HLS_SSR`, for both this design and the following design discussed in section 3.2.4. Three sets of 32 tap registers were used, two sets of which were used to carry out high-performance computation of the tap shift and sum. The third set was employed for the restoration of SSR values from the external, from-scratch SSR mechanism written previously to the project. 32 constant-type registers were used to store the coefficients, for maximum accessibility.

Tap Manipulation and Restoration

Two sets of tap registers were used in the core filtration mechanism for the purpose of increasing the HLS capability to parallelize the shifting and summation of the input ADC values. This is because internal loop dependencies occur when an array must read and write to itself on 32 different occasions, seen in the C++ code featured within listing 1.

Listing 1: The for loop responsible for the tap shift and summation in C++, using only one set of tap registers

```

for (short i = 31; i >= 0; i--) {

    // New ADC input introduced to first register
    if (i == 0) {
        tap_array[i] = ADC_input;
    }

    // Tap value shift
    else {
        tap_array[i] = tap_array[i-1];
    }
    // Sum of each tap occurs simultaneously
    sum += coefficients[i] * tap_array[i];
}

```

If this loop is modified so that there are two arrays that take turns being read from or written to, i.e. `tap_array0[i] = tap_array1[i-1]`; or vice versa, we can eliminate this dependency and achieve higher performance since all 32 sets reads and writes can occur at the same time, including the sum operations. When the `tlast` restoration occurs, we restore the temporary SSR registers to both tap arrays.

In order to communicate with and restore tap values from the external SSR, additional related ports were implemented to accept and validate the incoming SSR tap values. The timing of the tap restoration was important; the SSR block output the first stored ADC value after the 32nd input sample of the packet had been processed, and the final stored ADC value one clock cycle after `tlast` went high, so contingencies were put in place to accommodate this and make sure all taps were restored correctly. This is in contrast to the pedestal subtraction SSR output which restored both the median and accumulator values several clock cycles after the end of the packet, preventing any ambiguity in the restoration.

Performance and Resource Utilization Comparison

The maximised performance of the `fir_HLS` design required many different trial and error attempts at optimisations and creating more opportunities for parallelization of the most time-expensive function operations: namely, the tap summation and SSR restoration. The significant disparity in the resources featured in figures 18-19 reflect the increased registers and associated operations required to create a functional FIR design in HLS whilst overcoming the obstacles created by FELIX's required AXI stream behaviour. The LUT usage in the new HLS design is a 250% increase from the amount used by the previous block, with a non-negligible 0.23% increase in the total resource usage. The same goes for the FF usage, with an 102% increase from the previous algorithm's usage, and a notable 0.14% increase in total resource usage.

In contrast to with the pedestal subtraction algorithms, DSP usage is required in both the original and HLS blocks. However, DSP usage in the HDL block is 0.39% higher than the HLS design, in terms of total FPGA usage. This is the desired utilization, and effort were made to increase DSP48 usage in HLS using directives to no avail. This would have allowed savings of the resource cost in the LUT category. IO utilization is similar in both the HDL and HLS algorithms, due to the extra ports required for restoration.

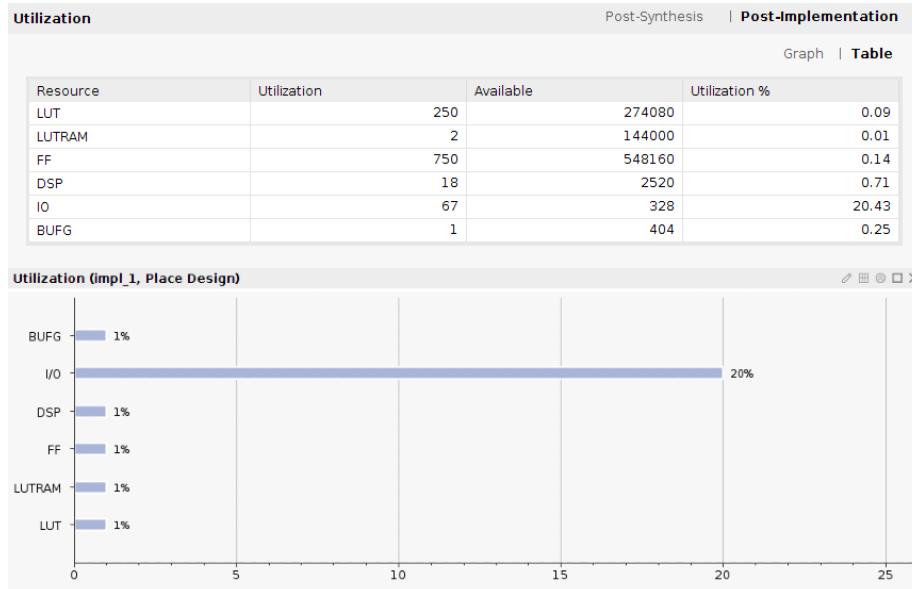


Figure 18: This figure details the resource statistics of the previously written FIR filter algorithm with the SSR mechanism not included, to allow for comparison to the analagous redesigned HLS module.

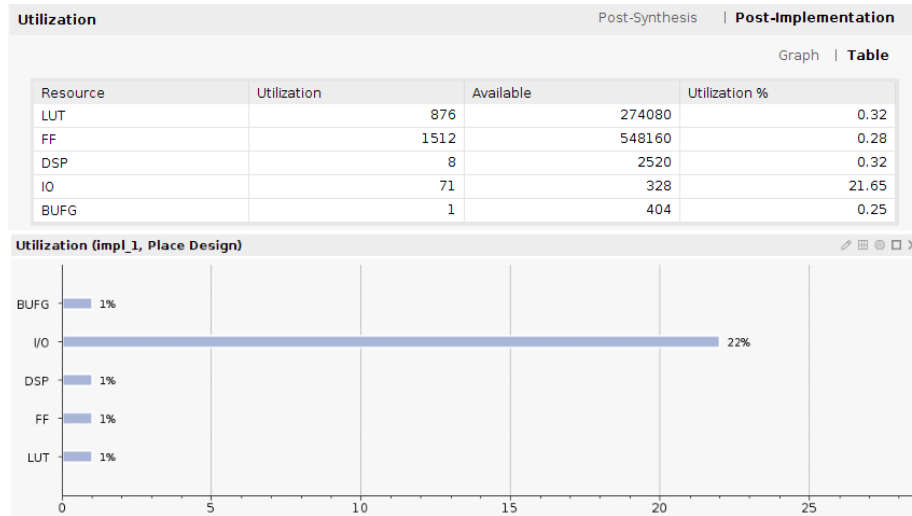


Figure 19: This figure contains the resource statistics for the FIR filter algorithm written in HLS, connected to a previously written external SSR mechanism that is not included in the statistics. The only resources being spent on the SSR are those allocated for accepting an storing the output values from the external SSR block.

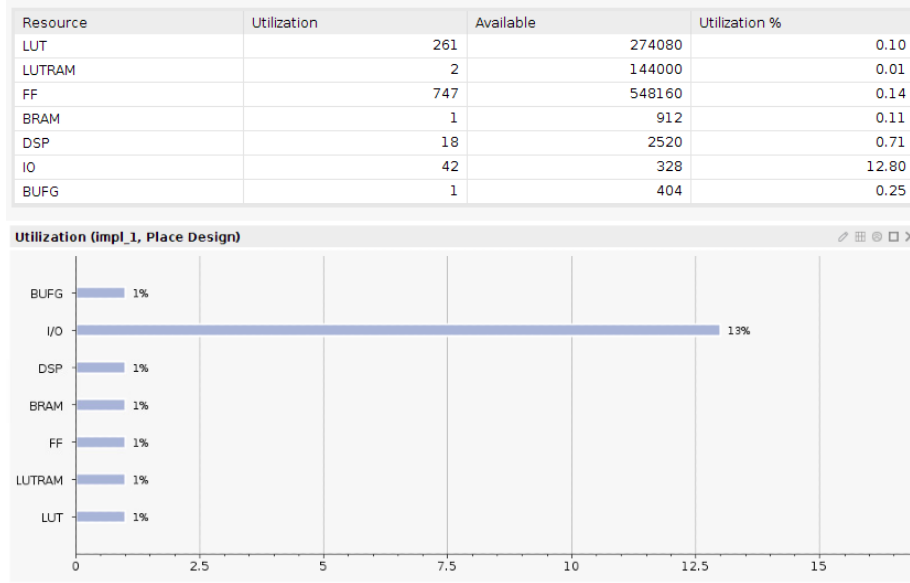


Figure 20: This figure shows the resource statistics for the previously built FIR algorithm with the SSR mechanism included in the usage. This is included to allow the comparison to the HLS FIR block with an in-built SSR mechanism.

3.2.4 FIR with Implemented SSR

The `fir_HLS_SSR` block employed a non-partitioned, two-dimensional array implemented as a BRAM to handle its own tap saving and restoring behaviour. Due to the access limitations of the BRAM component, tap restoration to the temporary SSR registers for the next channel occurs during the first 32 samples of each packet, i.e. one value restored per clock cycle. This is so that there are no read operations occurring during the final 32 samples, where the BRAM is written to by the incoming ADC values so that they can be restored the next time that the current channel receives a packet. In order for this block to reach the required performance, false dependency directives were applied to the temporary 32-register restoration array and the two-dimensional SSR array. Additionally, the same double-array tap shift method was used from section 3.2.3. This allowed full parallelization of SSR operations, and the design output was verified in the C/RTL cosimulation and Questasim simulations.

Performance and Resource Utilization Comparison

The `fir_HLS_SSR` implementation has the biggest disparity in resources compared to the original design, likely due to the reasons discussed within section 3.2.3, in addition to the framework used in the internal SSR. This demanded the storing and retrieving of far more values than `pedsub_HLS_SSR`. Through

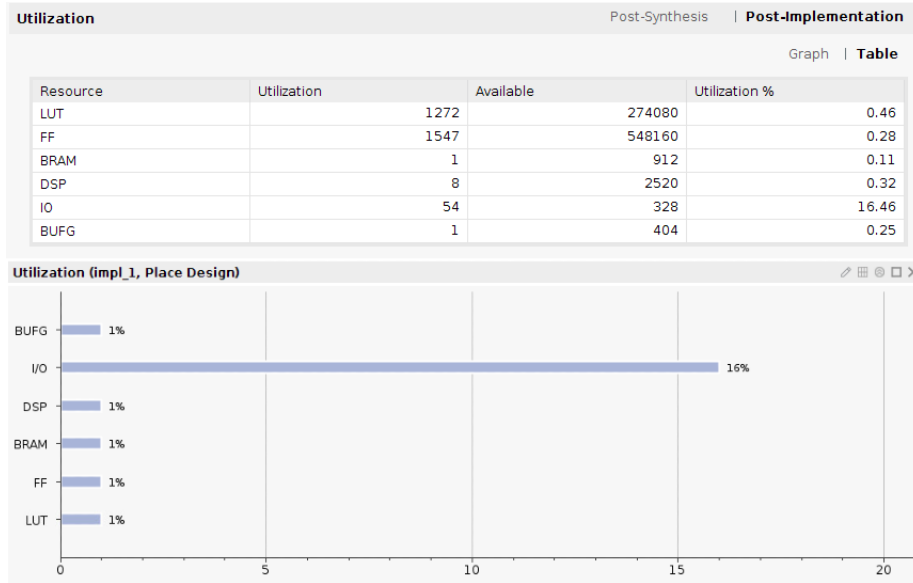


Figure 21: The above graphic demonstrates the resource statistics for the HLS-synthesized FIR algorithm with an in-built SSR memory and access mechanism, to enable the comparison to the version written from scratch in a HDL.

the analysis of figures 20 and 21 it can be deduced that the HLS block requires 387% more LUTs than the manual HDL design, with a 0.36% increase in usage of total LUTs available, the largest increase yet. FF usage in the new block came to a 107% increase from the previous design, again using 0.14% more of the total FFs available. Similar statistics comparisons emerge from section 3.2.3 regarding the remaining resources in this section, besides the use of one BRAM component in both blocks for the SSR. This tells us that the main difference between `fir_HLS` and `fir_HLS_SSR` is unsurprisingly the LUT usage.

4 Design Processs Conclusions

4.1 HLS and Manual HDL Resource Usage with Similar Performance

Given more time to investigate `axis` interface directives so that every design was not pushed to complete operations within 4 nanoseconds, insight could be provided into resource usage of pipelined HLS-designed blocks with the same latency as the original block. In this case, the efficiency of high-level synthesised implementations could be brought into question. Papers have been previously written that compare HLS and manual VHDL implementations for high energy physics, among other purposes, and one general conclusion is that HLS pro-

duces a less efficient implementation, but allows for faster development time and better design productivity [49–51]. Other papers discuss the tools provided by Vivado HLS as mature enough to consider as the first approach for designing FPGA algorithms, and sometimes provides similar, or even more efficient implementations when dealing with the re-use of certain resources [52, 53].

During the design of the firmware algorithms discussed in this report, once the user became more familiar with the HLS tools available, design productivity increased dramatically and functional IP blocks were implemented within increasingly shorter periods. However, this cannot be separated from the fact that AXI4 stream behaviour is far more easily implementable in manual HDL projects than it is in their HLS counterparts.

4.2 Simulation using Questasim

Arguably, Questasim is a very useful software for gauging the behaviour of HLS IP in a more realistic environment i.e. closer to the conditions the firmware would experience in a detector. However, the disparity between validated results in the HLS testbench environment and the complete malfunction of output signal drivers in the digital circuit can lead to roadblocks in the design process. Increased development productivity comes at a price: once the IP is synthesized in HLS it proves difficult to analyse the machine generated code. The Questasim waveform can create a window into the contributing drivers for a signal not driven properly, or exhibiting the wrong behaviour. Figure 22 provides a visualisation of the signal patterns for `pedsub.HLS.SSR`. Comparing the output to existing blocks also mapped in the waveform, or the model output text files for a given input file will often draw attention to the source of an issue. Timing-specific problems are best analysed in the signal waveform as it is the best place to spot misalignments of inputs and outputs. However, if this approach was unsuccessful then the remaining solution was to try a different method to implement the desired behaviour, or a simplified version of the current algorithm. This proved to be time-consuming and imprecise.

There were several points during development where simulations froze inexplicably when processing a certain component of the IP, and were fixed by arbitrary changes in the wrapper. Partially pipelined designs i.e. an average $II = 1$ but a maximum of $II = 2$ was one of several causes of said freezing. Other triggers of erroneous simulation runtime included mismanagement of the IP source hierarchy in the project configuration file, or the incorrect activation of certain blocks in the TPG chain. Finally, on occasion the output write file produced outputs that did not correlate with the waveform, generating questions as to how Questasim visualised outputs versus their actual readout. All of these issues were part of a long trial and error process. The customisability of manual HDL development would provide considerably more transparency into debugging when faced with the Questasim validation phase. HLS provides the edge in productivity during the previous design phases.

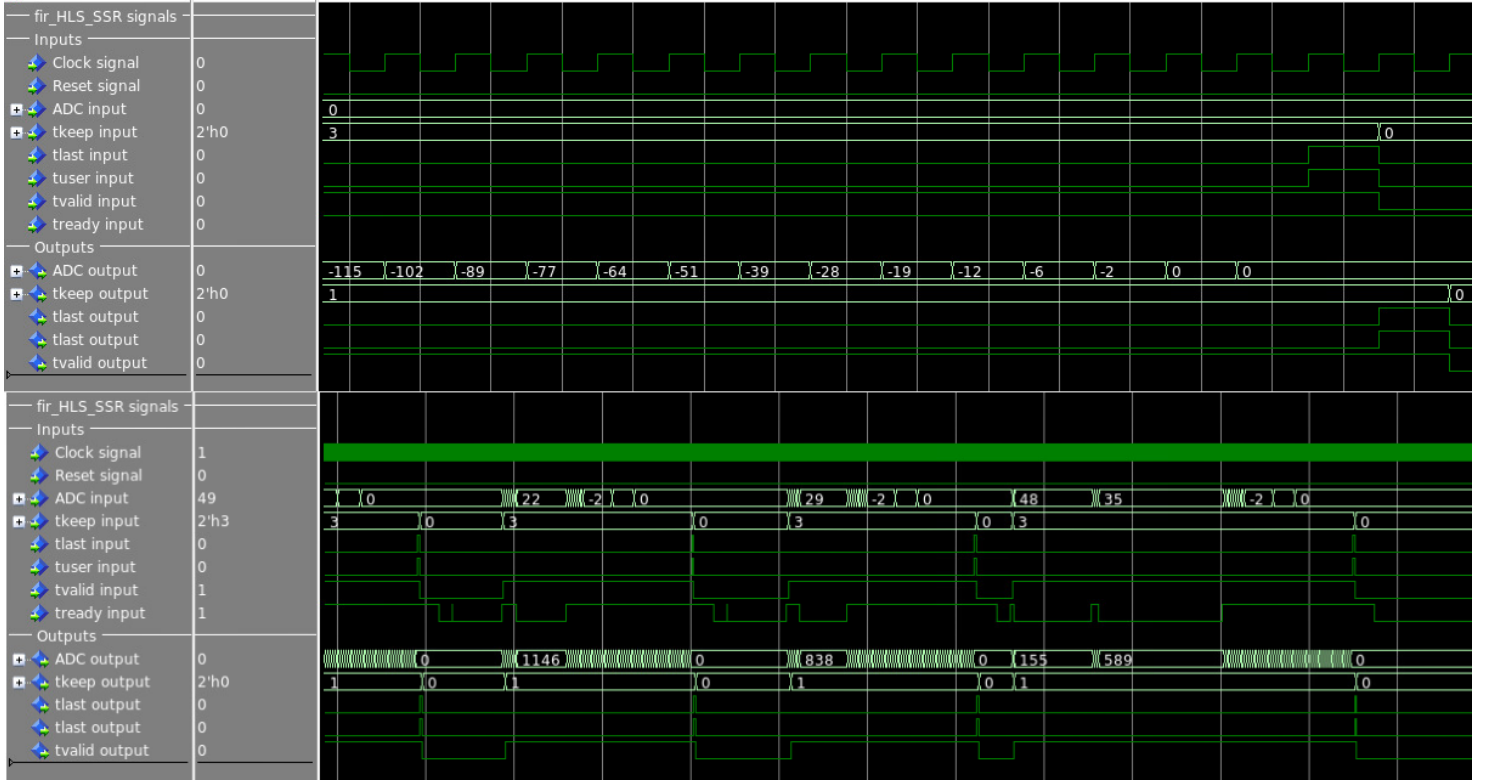


Figure 22: This waveform in this figure focusses on a slice of all input and output port signals for `fir_HLS_SSR` in the Questasim simulation environment. The image above features a zoom-in view on signal behaviour at the end of a singular packet, and the image below visualises several packets. Note the accessible decimal readout for ADC input and output, rapidly changing packet bunches during high `tvalid` and `tkeep` input signals and no change in the ADC value after a high `tlast` input, as described in section 3.1. The `tready` input signal oscillates randomly during the simulation, pausing block operations temporarily. This simulates sequential blocks sending back pressure due to limited throughput and too many input values. This waveform provides valuable information into the block operation and the clock timing of each AXI4 signal.

4.3 Final Thoughts

List because this part is unfinished.

- Chronology affects the operation of HLS blocks unless you use the inbuilt interface systems such as the axis directives
- The high-performance approach to achieving the required behaviour was sub optimal and resource inefficient, and could be improved by using the axis directives or finding another way of implementing back pressure behaviour. This is far easier to achieve in manual HDL.
- Debugging involves lots of trial and error, starting with simplistic designs so less can go wrong (due to lack of transparency into what is causing issues) in contrast to manual HDL debugging.
- The HLS synthesis and analysis GUI is very helpful to HDL firmware novices - timing, DRC's e.t.c. Applying directives and pragmas is very intuitive, allows for different implementations of the same source code
- HLS allows for higher design productivity but less efficient resource usage regardless of the performance. This isn't crippling to the overall utilization for a certain resource, in our case. Talk about overall comparison between original and HLS designs.
- Lots of unnecessary resources used to achieve latency = $\Pi = 1$, such as extra sets of array registers and partitioning of BRAM memories. However, it is impressive that this kind of throughput can be achieved for relatively complex top functions

5 Future of High Level Synthesis and the DUNE Experiment

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